DSA 210: Introduction to Data Science
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# AI and the Future of Entry-Level Jobs: Are Recent Graduates at Risk?

*An exploratory and statistical analysis of AI-related job demand, salary differences, education requirements, and early-career accessibility*

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# 1. Motivation

Artificial intelligence has become one of the most discussed topics in technology, education, and the job market. As a Computer Science student, I often hear that AI will create many new opportunities, but at the same time, I also hear concerns that it may make it harder for recent graduates to find entry-level jobs. This topic feels personally relevant because I am preparing to enter the job market in the near future, and I want to understand whether these concerns are supported by data or whether they are mostly based on assumptions.

One reason I chose this project is that entry-level jobs are very important for students and new graduates. These jobs are usually the first step into a career, helping people gain experience, improve their technical skills, and understand how the industry works. However, if AI-related roles increasingly require advanced skills, higher education, or previous work experience, recent graduates may face additional barriers when trying to start their careers. This made me curious about whether AI is really opening doors for young professionals or making the job market more competitive and difficult to access.

This project focuses on AI-related job market data to explore three main questions: whether job demand differs by experience level, whether salary and AI salary premiums differ across experience levels, and whether education requirements are related to experience level. These questions are important because they connect directly to concerns that many university students have: “Are there enough entry-level opportunities?”, “Do AI skills increase salary equally for everyone?”, and “Do I need a formal degree or advanced qualifications to enter this field?”

Overall, I am working on this project because it connects a real-world issue with data analysis. Instead of only discussing AI’s impact on jobs in a general way, this project uses actual job market variables such as demand score, salary, AI salary premium, education requirements, required skills, and experience level. By analyzing these variables, I aim to better understand how AI is affecting early-career opportunities and what this may mean for students and recent graduates entering the technology field.

As a student, I wanted this project to answer a question that is not only academically interesting, but also personally and practically important for my future career decisions.

## 2. Datasets

The data used in this project comes from a Kaggle AI job market dataset. I chose this dataset because it directly includes the variables needed to study entry-level accessibility in AI-related jobs, such as experience level, demand score, salary, AI salary premium, education requirements, and required skills.

### 2.1 Dataset Source

The original dataset was downloaded from Kaggle and then cleaned for this project. The final dataset used in the analysis is: data/ai_jobs_market_cleaned.csv
Each row represents an AI-related job record, and the columns describe job characteristics such as salary, demand, education expectations, required skills, and experience level.

### 2.2 Variables Used

The main variables used in the analysis are:

| |
|---|
| demand_score |
| experience_level |
| annual_salary_usd |
| ai_salary_premium_pct |
| education_required |
| required_skills |

`experience_level` is the main comparison variable because the project focuses on differences between entry-level, mid-level, senior, and lead roles. `demand_score` is used to examine job demand, while `annual_salary_usd` and `ai_salary_premium_pct` are used to analyze salary differences. `education_required` and `required_skills` help evaluate whether AI-related jobs create additional barriers for early-career candidates.

## 3. Data Collection & Preparation

The dataset was collected by downloading the original AI job market data from Kaggle. After downloading it, I cleaned and organized the data in Jupyter Notebook using Python. The goal of this step was to create one consistent dataset that could be used across the whole project.

### 3.1 Initial Data Acquisition

The raw dataset was first imported into Jupyter Notebook with `pandas`. I checked the column names, data types, missing values, and the general structure of the dataset. This helped me understand which variables were useful for my research questions and which ones were not necessary for the final analysis.

### 3.2 Data Cleaning Steps

The cleaning process focused on keeping the project simple and consistent. I selected the variables most relevant to the research questions. Then, I removed rows with missing values in these key columns. This was important because the hypothesis tests and machine learning models require complete and consistent data. I also made sure that numerical variables such as salary, demand score, and AI salary premium were usable for statistical analysis and modeling.

### 3.3 Final Dataset

After cleaning, the final dataset was saved in the `data` folder and used in all notebooks.

## 4. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted before hypothesis testing to understand the structure of the dataset and identify initial patterns. In this stage, I examined the distribution of experience levels, salary, AI salary premium, demand score, education requirements, and correlations between numerical variables.

## 4.1 Dataset Overview

The first step was to inspect the cleaned dataset by checking its shape, column names, data types, and summary statistics. This helped confirm that the data was correctly loaded and ready for analysis. It also gave a general understanding of which variables were numerical and which were categorical.

## 4.2 Experience Level Distribution

I first visualized the distribution of `experience_level` to see how job listings were spread across different career stages. This was important because the whole project compares entry-level, mid-level, senior, and lead roles. Checking this distribution helped confirm whether there were enough observations in each group before moving to hypothesis testing.

## 4.3 Salary Analysis

The salary analysis focused on `annual_salary_usd`. I first examined the overall salary distribution to understand whether salaries were normally distributed or skewed. Then, I compared salary distributions across experience levels using a boxplot. This visualization showed whether salaries tend to increase as experience level increases, which directly supports the second hypothesis about salary inequality.

## 4.4 AI Salary Premium Analysis

I also analyzed `ai_salary_premium_pct` across experience levels. This was important because the project does not only ask whether senior roles earn higher total salaries, but also whether AI-related salary premiums differ across experience levels. This distinction matters because higher total salary and higher AI-specific premium are not necessarily the same thing.

## 4.5 Demand Score Analysis

To understand job accessibility, I compared `demand_score` across experience levels. This analysis helped explore whether demand is concentrated more strongly in experienced roles or whether entry-level AI jobs also appear in the market. This visualization directly supported the first hypothesis, which tests whether demand score differs across experience levels.

### 4.6 Education Requirement Analysis

For education requirements, I created a crosstab between `experience_level` and `education_required`, and also visualized education requirements by experience level. This helped show how formal education expectations are distributed across job levels. Since degree requirements can create barriers for recent graduates, this analysis directly connects to the third hypothesis.

### 4.7 Correlation Analysis

Finally, I created a correlation matrix for the main numerical variables: `annual_salary_usd`, `ai_salary_premium_pct`, `demand_score`, and `years_of_experience`. This helped examine whether these variables move together. For example, it allowed me to see whether salary, AI salary premium, demand, and years of experience had positive or negative relationships.

Overall, the EDA showed that salary differences across experience levels were visually clear, while demand score, AI salary premium, and education requirements needed formal hypothesis testing for stronger interpretation. The EDA stage therefore helped guide the statistical tests and machine learning analysis that followed.

## 5. Hypothesis Testing

After completing the exploratory data analysis, I used formal hypothesis testing to check whether the patterns observed in the visualizations were statistically meaningful. This step was important because visual differences can sometimes appear in graphs even when they are not statistically significant.

The hypothesis testing section focuses on three main questions: demand, salary, and education requirements across experience levels.

## 5.1 Research Questions & Hypotheses

### Hypothesis 1: Demand and Experience Level

The first hypothesis examines whether AI job demand differs across experience levels.

**Research Question:**
Does demand score differ across experience levels?

**H0:** Mean demand score is equal across experience levels.
**H1:** Mean demand score differs across experience levels.

This hypothesis was tested using Welch's ANOVA because the goal was to compare the mean `demand_score` across multiple experience-level groups.

### Hypothesis 2: Salary Inequality

The second hypothesis examines whether salary-related outcomes differ across experience levels. I tested this in two parts because total annual salary and AI salary premium measure different things.

**Part A: Annual Salary**

**Research Question:**
Does annual salary differ across experience levels?

**H0:** Mean annual salary is equal across experience levels.
**H1:** Mean annual salary differs across experience levels.

**Part B: AI Salary Premium**

**Research Question:**
Does AI salary premium differ across experience levels?

**H0:** Mean AI salary premium is equal across experience levels.
**H1:** Mean AI salary premium differs across experience levels.

Both parts were tested using Welch's ANOVA because the goal was to compare numerical variables across multiple experience-level groups.

### Hypothesis 3: Education Requirements

The third hypothesis examines whether education requirements are related to experience level.

**Research Question:**
Are education requirements associated with experience level?

**H0:** Education requirements are independent of experience level.
**H1:** Education requirements are associated with experience level.

This hypothesis was tested using a Chi-square test of independence because both `education_required` and `experience_level` are categorical variables.

## 5.2 Methodology

For numerical comparisons, I used Welch's ANOVA instead of a standard ANOVA because it is more reliable when group variances may not be equal. This was appropriate for comparing `demand_score`, `annual_salary_usd`, and `ai_salary_premium_pct` across experience levels.

For the categorical education requirement analysis, I used a Chi-square test of independence. This test checks whether two categorical variables are statistically associated.

For all hypothesis tests, I used a significance level of: $\alpha = 0.05$.

If the p-value was below 0.05, I rejected the null hypothesis. If the p-value was greater than or equal to 0.05, I failed to reject the null hypothesis.

## 5.3 Results

### Hypothesis 1 Result

For demand score, the Welch's ANOVA test produced a p-value of approximately: p-value = 0.2956.

Since this p-value is greater than 0.05, I failed to reject the null hypothesis. This means that there is not enough statistical evidence to conclude that demand score differs significantly across experience levels.

Although the EDA showed some small differences in demand score between groups, these differences were not statistically significant.

### Hypothesis 2 Result

For annual salary, the Welch's ANOVA test produced a very small p-value: p-value = 7.7359e-102.

Since this value is far below 0.05, I rejected the null hypothesis. This means that annual salary differs significantly across experience levels. In other words, experience level is strongly related to salary.

For AI salary premium, the p-value was: p-value = 0.5791.

Since this value is greater than 0.05, I failed to reject the null hypothesis. This means that there is not enough evidence to conclude that AI salary premium differs significantly across experience levels.

This result is especially interesting because annual salary clearly changes with experience level, but the AI-specific salary premium does not show the same statistically significant pattern.

### Hypothesis 3 Result

For education requirements, the Chi-square test produced the following result:

Chi-square statistic = 14.7168

Degrees of freedom = 12

p-value = 0.2573

Since the p-value is greater than 0.05, I failed to reject the null hypothesis. This means that there is not enough statistical evidence to conclude that education requirements are associated with experience level in this dataset.

Therefore, although some AI-related jobs may require formal education, the test does not show a statistically significant relationship between education requirements and experience level.

## 5.4 Interpretation

The hypothesis testing results show that not all differences observed in the data are statistically significant. The strongest finding is related to salary. Annual salary differs clearly across experience levels, which supports the expectation that senior and lead roles generally earn more than entry-level roles.

However, the result for AI salary premium is more surprising. Even though senior roles have higher overall salaries, the AI-specific salary premium does not significantly differ across experience levels. This suggests that AI-related skills may provide value across different career stages, not only for senior employees.

The demand score result also shows that demand is not significantly different across experience levels. This is important because it suggests that entry-level AI-related roles may not be completely excluded from demand in the job market. However, demand alone does not guarantee accessibility, because job requirements and skills still matter.

Finally, the education requirement test did not find a statistically significant association between education requirement and experience level. This means that the dataset does not provide strong evidence that formal education requirements differ by job level. From the perspective of recent graduates, this result is less negative than expected, because it does not statistically confirm that entry-level AI jobs face uniquely higher formal degree barriers.

Overall, the hypothesis testing results suggest that experience level strongly affects salary, but its relationship with demand, AI salary premium, and education requirements is less clear. This makes the project more interesting because it shows that the impact of AI on early-career opportunities is not one-dimensional. The main challenge for recent graduates may not be only demand or formal education, but the broader combination of salary structures, required skills, and competition in the AI job market.

## 6. Machine Learning Modelling

In the machine learning part of the project, I used the cleaned dataset to build predictive models for two outcomes: job experience level and annual salary. The purpose was not to build a perfect prediction system, but to understand whether the available job market features could capture patterns related to accessibility and salary in AI-related jobs.

### 6.1 Problem Setup

I designed two machine learning tasks. The first task was a classification problem where the target variable was `experience_level`. The goal was to predict whether a job belongs to an entry-level, mid-level, senior, or lead category. The second task was a regression problem where the target variable was `annual_salary_usd`. The goal was to predict salary using the available job-related features.

For both tasks, the data was split into training, validation, and test sets. The training set was used to fit the models, the validation set was used for comparison and tuning, and the test set was used for final evaluation.

### 6.2 Classification: Predicting Experience Level

For classification, I used Logistic Regression as a baseline model and Gradient Boosting Classifier as a more flexible model. The input features included variables such as demand score, annual salary, AI salary premium, education requirement, and required skills.

The models were evaluated using accuracy, precision, recall, and F1-score. I did not rely only on accuracy because experience-level prediction can be uneven across different classes. F1-score was especially useful because it considers both precision and recall.

The tuned Gradient Boosting model achieved a test F1-score of approximately **0.376**. This shows that the classification task was difficult. The model learned some patterns, but it could not strongly predict experience level. One possible reason is that experience level may depend on job responsibilities, company expectations, and task complexity, which are not fully represented in the dataset.

## 6.3 Regression: Predicting Annual Salary

For regression, I used Linear Regression as a baseline model and Gradient Boosting Regressor as a more flexible model. The target variable was `annual_salary_usd`.

The regression models were evaluated using MAE, RMSE, and $R^2$. These metrics are more appropriate than accuracy because salary is a numerical variable. MAE and RMSE show the average size of prediction errors, while $R^2$ shows how much of the salary variation is explained by the model.

The tuned Gradient Boosting Regressor achieved a test RMSE of approximately **51,802** and a test $R^2$ score of approximately **0.345**. This means the model could explain part of the variation in salary, but salary prediction remained challenging.

## 6.4 Hyperparameter Tuning with GridSearchCV

To improve model performance, I used GridSearchCV for hyperparameter tuning. GridSearchCV tests different parameter combinations and selects the best model based on cross-validation performance.

For the classification task, GridSearchCV selected the following Gradient Boosting Classifier parameters: learning_rate = 0.05, max_depth = 3, n_estimators = 100.

The best cross-validation F1-score was approximately **0.391**.

For the regression task, GridSearchCV selected the following Gradient Boosting Regressor parameters: learning_rate = 0.1, max_depth = 3, n_estimators = 100.

The best cross-validation RMSE was approximately **49,560**.

Using GridSearchCV made the model selection process more systematic because the model settings were chosen based on validation performance instead of manual guessing.

### 6.5 Model Comparison and Interpretation

The classification models were compared using accuracy, precision, recall, and F1-score. The results showed that predicting experience level was difficult. The tuned model performed better on the training set than on the validation and test sets, suggesting some overfitting. However, the validation and test results were close to each other, meaning the model's performance on unseen data was relatively consistent.

The regression models were compared using MAE, RMSE, and $R^2$. The tuned Gradient Boosting Regressor performed better on the training set than on the validation and test sets, which also suggests some overfitting. This limitation is important because the model may learn patterns specific to the training data that do not fully generalize to new job postings.

Overall, the machine learning results show that salary prediction was more informative than experience-level classification. The models could capture some patterns in the job market, but the available features were not enough to fully explain the complexity of AI job roles. This supports one of the main ideas of the project: AI job accessibility and salary outcomes are influenced by multiple factors, and simple dataset features can only explain part of the picture.

## 7. Key Findings

The analysis showed that annual salary is the clearest difference across experience levels. Senior and lead roles generally have higher salaries than entry-level roles, and this difference was statistically significant.

However, AI salary premium did not significantly differ across experience levels. This was an interesting result because it suggests that AI-related skills may provide value across different career stages, not only for senior employees.

The analysis also showed that demand score did not significantly differ across experience levels. This means that the dataset does not provide strong evidence that AI job demand is concentrated only in higher-level roles.

For education requirements, the Chi-square test showed no statistically significant association between education requirement and experience level. Therefore, the data does not strongly support the idea that entry-level AI jobs have uniquely different formal education requirements.

Finally, the machine learning results showed that predicting salary was more informative than predicting experience level. Experience level was difficult to predict, which suggests that job level may depend on factors not fully captured in the dataset.

## 8. Key Visualizations

This section presents the most important visualizations from the EDA and machine learning stages. These figures were selected because they directly support the project's main questions about demand, salary, education requirements, and model performance.

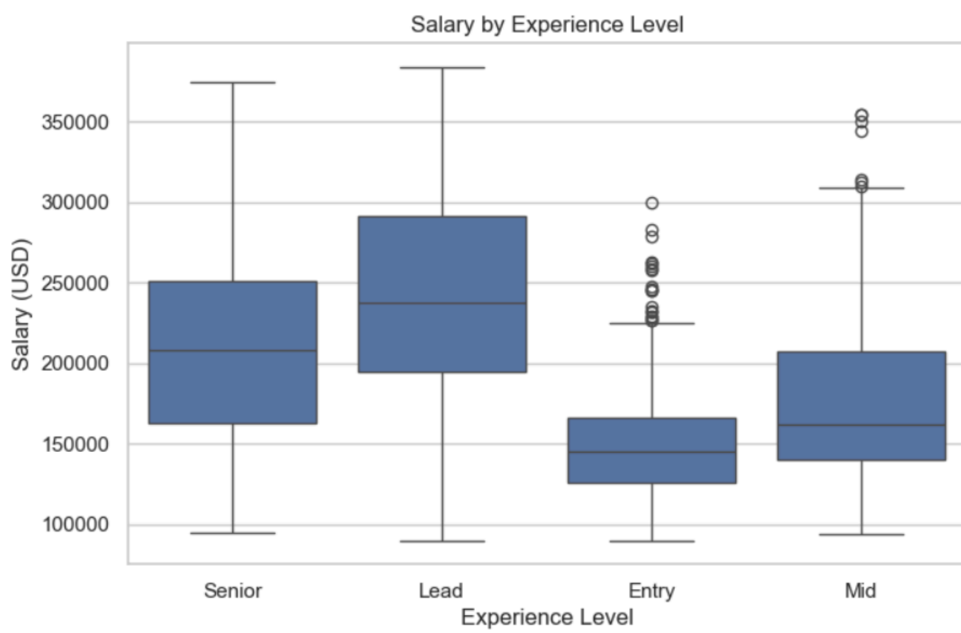


*Figure 1. Annual salary by experience level.*

This figure shows that higher experience levels generally have higher salary ranges, which supports the hypothesis testing result for annual salary.


*Figure 2. AI salary premium by experience level.*

This figure helps compare whether the additional salary benefit related to AI differs across experience levels.

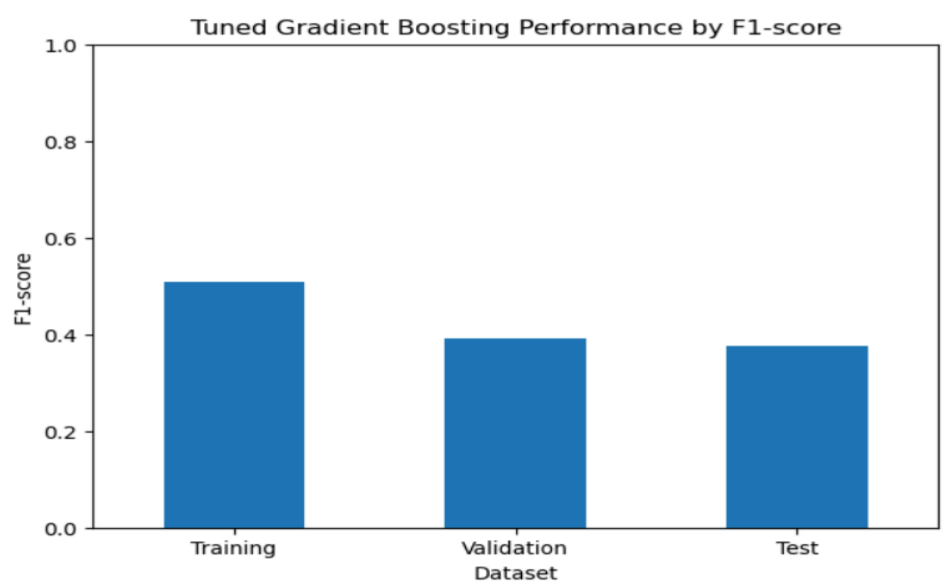


*Figure 3. Tuned Gradient Boosting Classifier Performance by F1-score*

This figure compares training, validation, and test F1-scores for the tuned Gradient Boosting Classifier. The model performs better on training data than unseen data, showing that predicting experience level is challenging.

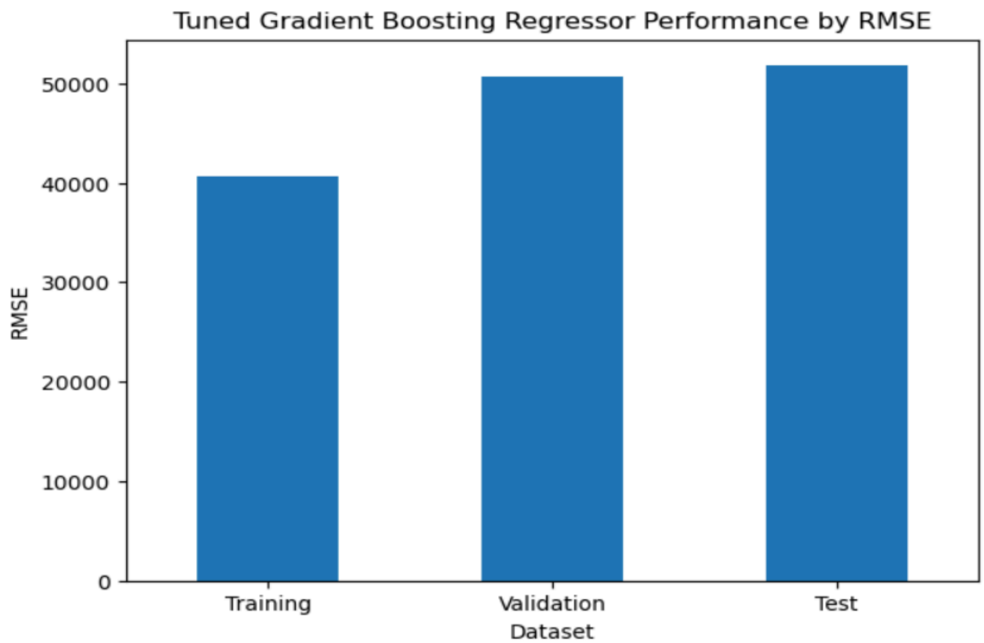


*Figure 4. Tuned Gradient Boosting Regressor Performance by RMSE*

This figure compares training, validation, and test RMSE values for the tuned Gradient Boosting Regressor. The validation and test errors are close, but both are higher than the training error, suggesting some overfitting.

## 9. Limitations & Future Work

One limitation of this project is that the analysis uses one cleaned Kaggle dataset. Because of this, the findings may not represent the entire AI job market. The dataset may also reflect the structure and assumptions of the source it came from.

Another limitation is that some important job factors are not included in the data. For example, company reputation, location, job responsibilities, interview difficulty, portfolio quality, and candidate background could also affect job accessibility and salary.

The machine learning models also showed some signs of overfitting, especially in the regression task. The training performance was stronger than the validation and test performance, which means the model may have learned some patterns specific to the training data.

In future work, I would like to use larger and more diverse job posting datasets from multiple sources. It would also be useful to include text analysis on job descriptions to better understand required skills, responsibilities, and entry-level accessibility. More advanced models and wider hyperparameter tuning could also improve prediction performance.

## 10. Academic Integrity Statement

I used AI tools, specifically ChatGPT, as a support tool during some stages of the project. The AI assistance was used for guidance, organization, debugging support, and improving the clarity of written explanations. I did not use AI to replace my own analysis or decision-making. Instead, I used it to better understand errors, organize the project structure, and improve the wording of the report.

The specific types of AI assistance used include:
- organizing the project repository according to the instructor’s feedback;
- rewriting and clarifying parts of the README and proposal document;
- debugging Python and Jupyter Notebook errors;
- improving the explanation of hypothesis testing results;
- suggesting how to structure the final report;
- improving the clarity and grammar of written sections.

## 11. Conclusion

This project explored whether AI-related jobs are accessible for entry-level candidates and recent graduates. The results showed that salary differs strongly across experience levels, but demand score, AI salary premium, and education requirements did not show statistically significant differences across experience levels. The most important finding is that AI job accessibility is not explained by one single factor. Salary clearly increases with experience, but AI-specific salary premium and education requirements show a more complex pattern. This suggests that recent graduates may still have opportunities in AI-related fields, but they need to be aware of skill expectations and competition.

Overall, this project helped me better understand a real concern that many Computer Science students have: whether AI is creating opportunities or making the job market harder to enter.

The results suggest that the situation is mixed. AI-related jobs may offer opportunities for early-career candidates, but salary growth and role progression still depend strongly on experience and broader job market factors.